

Prevalence of celiac disease in at-risk and not-at-risk groups in the United States: a large multicenter study.

BACKGROUND: Celiac disease (CD) is an immune-mediated enteropathic condition triggered in genetically susceptible individuals by the ingestion of gluten. Although common in Europe, CD is thought to be rare in the United States, where there are no large epidemiologic studies of its prevalence. The aim of this study was to determine the prevalence of CD in at-risk and not-at-risk groups in the United States. **METHODS:** Serum antigliadin antibodies and anti-endomysial antibodies (EMA) were measured. In EMA-positive subjects, human tissue transglutaminase IgA antibodies and CD-associated human leukocyte antigen DQ2/DQ8 haplotypes were determined. Intestinal biopsy was recommended and performed whenever possible for all EMA-positive subjects. A total of 13 145 subjects were screened: 4508 first-degree and 1275 second-degree relatives of patients with biopsy-proven CD, 3236 symptomatic patients (with either gastrointestinal symptoms or a disorder associated with CD), and 4126 not-at-risk individuals. **RESULTS:** In at-risk groups, the prevalence of CD was 1:22 in first-degree relatives, 1:39 in second-degree relatives, and 1:56 in symptomatic patients. The overall prevalence of CD in not-at-risk groups was 1:133. All the EMA-positive subjects who underwent intestinal biopsy had lesions consistent with CD. **CONCLUSIONS:** Our results suggest that CD occurs frequently not only in patients with gastrointestinal symptoms, but also in first- and second-degree relatives and patients with numerous common disorders even in the absence of gastrointestinal symptoms. The prevalence of CD in symptomatic patients and not-at-risk subjects was similar to that reported in Europe. Celiac disease appears to be a more common but neglected disorder than has generally been recognized in the United States.